



Sony DPT-RP1 tablet

Sony is revamping their DPT-S1 13inch digital paper tablet with the new DPT-RP1 with an improved screen resolution. <http://digit.in/DPTRP1>



ASUS ZenScreen

After much delays, ASUS has finally a selling date of their \$249 portable USB-C monitor, the ASUS ZenScreen. <http://digit.in/ZenScreen>



Map of Babylon: Another contender for the earliest one out there

cause of the straight lines of latitudes and longitudes.

In the 18th century, mathematics and astronomy, particularly the work of Sir Isaac Newton, gave a boost to the method of determining longitude to the degree. Eventually, developments in lithographic and photochemical technologies led to maps with greater levels of detail being printed. But the next true revolution in mapping came in the 20th century.

A WEAPON

Most of early 20th century, advances in printing and photographic technology allowed for physical maps to become more accurate. But as the 1950s passed, computer based mapping software saw a rise in development, especially due to progress in nuclear weapon research. After Russia launched the Sputnik 1



Sputnik 1 can effectively be given credit for kickstarting modern mapping technology

in 1957, two American physicists at the Johns Hopkins Applied Physics Laboratory (APL), decided to monitor its radio transmissions. Thanks to Doppler effect, they realised quite soon that they could track the location of the satellite along their orbit. The very next problem they got working on was the inverse - tracking the location of the user if that of the satellite is known. This led to the development of the TRANSIT system, the first satellite navigation system which was used by the US Navy and tested successfully in 1960.

The next decade saw multiple systems like the Project 621B, Transit, Timatian, SECOR use different approaches to fix the exact location problem. While none of them entirely succeeded, a superior system was eventually developed using the best technology from each of these, resulting in the Navstar-GPS system. Keep in mind that all of this was restricted military technology, which was about to change soon due to a tragic event.

A TECHNOLOGY

After a Korean Air Lines Boeing 747 was shot down for straying into USSR airspace in 1983, US President Ronald Reagan issued a directive instructing GPS to be made available for public use. For quite a few years, the highest quality signal was still reserved for military use while the civilian signal was subjected to 'selective availability' or SA.

Ground based navigational systems that were being used so far by civilian services had poorer accuracy and greater costs involved. While this could be solved with GPS, the S.A signal was not good enough for the same. The United States military received multiple requests over the years from the Federal Aviation Administration (FAA), United States Coast Guard (USCG) and United States Department of Transportation (DOT) to remove the restrictions of S.A but it was refused on the grounds of security.

The 1980s and 90s saw the gradual rise of Differential GPS (DGPS), which increased the accuracy of the S.A signal

by taking into account the offset in the poorer signal, along with efforts by the US Coast Guard to make high quality signals available across the US. Eventually, most GPS vendors were offering receivers capable of using DGPS signals. By the mid-90s, DGPS was being used enough to render the S.A signal's purpose ineffective in the US. Additionally, during the Gulf War, the US forces had to use a large number of civilian receivers. These factors saw to the removal of the S.A restriction with an executive order from President Bill Clinton in 2000. By then, DGPS had evolved to provide more accuracy than a non-SA GPS signal, to the point where the best implementations offer an accuracy of up to 10cm.

Around the same time, Russian President Vladimir Putin paid special attention into reviving GLONASS, the Russian equivalent of GPS which had followed a similar trajectory to GPS until it fell into disrepair at a time of Russian economic crisis. By 2006, the signal was available for civilian use and by 2007, all restrictions were lifted.

A WAY OF LIFE

Over the next few years, commercial usage was heavily promoted for both GPS as well as GLONASS. The Russian government even implemented import duties specifically targeted at GPS devices to generate international interest into the GLONASS system. Both systems have seen technological upgrades and are due for more upgrades in the coming years.

Thanks to smartphones and now, cars with navigational capabilities, maps are a part of our lives in a much more integral way than we would have imagined even 10 years ago. Somewhere, along the way, we have developed a habit to not rely on our own sense of direction. Instead, listening to a computerised voice dictating your very steps seems to be a highly reliable way of finding the correct path. Cartography and navigation, as a field of technology, truly reflects the transition to an absolutely technology enabled evolution of humanity. 